SALES FORECASTING USING A STACKED ENSEMBLE MODEL

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***Abstract*— Zimbabwean retail businesses face a challenging economic landscape characterized by volatility and a heavy reliance on imports. Intuition-based planning methods prevalent in the sector often lead to significant losses due to overstocking or understocking. I propose a sales forecasting system that uses a stacked ensemble machine learning model to forecast future sales to address this challenge. The system leverages historical sales data to identify trends, seasonality, and patterns. Stacked ensemble models combine the strengths of multiple models, leading to improved generalizability and robustness. By providing accurate sales predictions, this system empowers businesses to anticipate demand, identify growth opportunities, and mitigate risks associated with economic volatility. This research contributes to the field of sales forecasting by demonstrating the effectiveness of stacked ensemble models in a real-world, economically volatile environment, particularly for Zimbabwean retail businesses.**

Keywords: sales forecasting, ensemble model, stacking, performance metrics

I. INTRODUCTION

The Zimbabwean retail industry despite its huge profitability in the country relies on instinct based decision making to plan market decisions to use. Despite its gross usage, it has its perks considering the currently volatile economic environment in the country which brings about unpredictability in the flow of the economic scale. As a result, many retail businesses suffer huge losses and even go bankrupt. To address this issue, this paper proposes a sales forecasting system using a stacked ensemble model to predict future sales based on historical sales data and the prevailing market and economic conditions. The ensemble models are trained on historical sales and data and the predictions from each of the algorithms are combined through stacking to produce a more robust forecasting model. The model is then integrated to a web application for interactive usability by the user. The web application then accepts user input for the current sales and features that affect sales then the system processes the input and forecasts future values for the specified duration.

Sales forecasting plays a critical role in the success of any retail business, allowing for informed decision-making regarding inventory management and marketing strategies [1]. Traditional forecasting methods based on intuition are inefficient in capturing complex relationships in sales data particularly in economic volatile markets thereby providing less accurate and less reliable predictions [12]. This is especially true for the Zimbabwean retail sector which faces challenges like import dependency and fluctuating exchange rates [3]. To address these limitations, advancements in machine learning techniques offer promising solutions.

Ensemble learning techniques which combine forecasts from multiple algorithms have emerged as powerful tools for improving the accuracy of sales forecasts [4]. Stacking which involves training a meta learner model on the forecasts from other base algorithms as new features has emerged as a very powerful method to improve robustness of the forecasts [5]. This approach leverages the strengths of various models to potentially capture diverse patterns within data and mitigate weaknesses of individual models [5].

In sales forecasting, ensemble models and stacking have gained popularity for their ability to generate more accurate and robust forecasts. By integrating various forecasting algorithms such as time series and regression, ensemble models can leverage the strengths of each approach to capture different aspects of the sales data.

Considering these developments, this paper proposes a sales forecasting system that uses a stacked ensemble model to forecast future sales and future sales trends accurately leveraging the power of ensemble models and the robustness of the models after stacking them.st. The web platform will provide an interactive platform for users to get valuable insights into the future using the power of machine learning which will help in effective decision making, formulating effective marketing strategies and effective inventory management.

II. PROBLEM STATEMENT

The Zimbabwean retail industry, a significant contributor to the national economy, suffers from substantial financial losses due to inefficiencies in inventory management and marketing strategies. These inefficiencies largely stem from a reliance on intuition-based decision making in a highly economically volatile environment. Traditional planning methods fail to account for critical factors like import dependency, fluctuating exchange rates, inflation, complex relationships in data and seasonal trends. Failure to account for these factors results in

overstocking which leads to global losses of $471.9 billion [13] or understocking with a global cost $34.1 billion [13], hindering profitability.

This highlights the critical need for a robust and accurate sales forecasting system specifically tailored to the Zimbabwean retail sector. Such a system would empower businesses to make informed decisions, mitigate risks associated with economic volatility, and achieve sustainable growth.

III. RELATED WORK

[8] presented a hybrid forecasting model combining the Holt-Winters and the Support Vector Machine algorithms to forecast the demand of spring onion seeds. The forecasting was based on dynamic factors including historical sales, seed inventory, price and weather [8]. The use of hybrid forecasting method led to more accurate predictions [1]. The model was designed to combine the strengths of the Holt-Winters algorithm and Support Vector Machines. The hybrid model was trained on a dataset of historical spring onion seed demand to learn to capture the patterns and relationships between input variables and the demand for spring onion seeds. An external factor was also added as a feature that would affect the demand of the spring onion seeds having a long-term influence on the demand which was the minimum absolute temperature. The performance metric used was the forecasting error which is the difference between the actual and the forecasted demand [8].

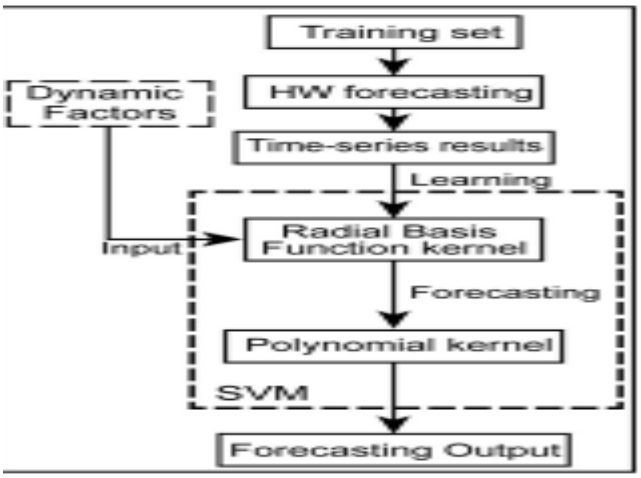

Figure 1: The control flow diagram for the training of the Hybrid model using Holt-Winters and Support Vector Machines for spring onion seed demand forecasting.

Dr. K. Alice Et al. (2023) carried out a forecast on the Walmart dataset using XGBoost, LGBM and Catboost algorithms in order to compare their performances.

The implementation of a hybrid model using all three of the algorithms using a voting regressor was also carried out to combine the strengths of all the algorithms and help remove biases of individual models. The XGBoost algorithm was chosen because it is more flexible, scalable and could be customized to fit the best hyperparameters. LGBM was chosen because it was the best in terms of memory efficiency overall efficiency. Catboost was expected to have the best results overally. The performance metrics used included the MAE, RMSE and R-squared. The Voting Regressor was evaluated using the MAE.

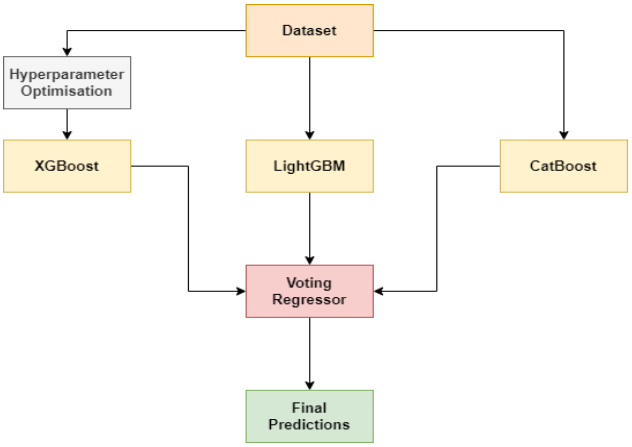

Figure 2: This shows the architecture of the hybrid ensemble model for the Walmart dataset.

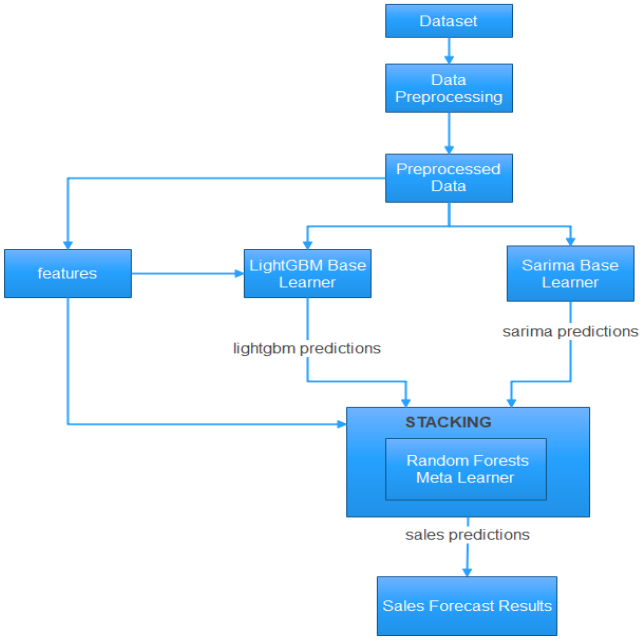

Figure 3: This shows the architecture of the proposed stacked ensemble model.

IV. SOLUTION

To eradicate the problems faced by the retail industry in Zimbabwe in terms of making informed decisions on market strategies and improve their competitiveness in an economically

volatile environment, I propose the following objectives:

- To develop a web-based sales forecasting system that predicts future sales by utilizing a machine learning model trained on historical data.
- To develop a web-based sales forecasting system that predicts sales trends over a period of time using historical data.
- To develop a web-based sales forecasting system that analyzes sales data.

A. Features of the System

The proposed system will have the following features:

1. Web Dashboard platform: The system will have a web portal that would allow users to access the functionalities of the system.
2. Sales forecasting: The system will allow users to input forecasting details of products and generate future sales values and trends according to the time period they require.
3. Sales Analysis: The system will provide a sales analysis feature that will allow users to analyze various aspects of their sales information.

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B. Benefits of the System

The proposed system offers the following benefits:

1. Enhanced decision making: Sales forecasts generated by the system will allow businesses to make informed decisions on marketing strategies based on future insight provided by the system.
2. Improved efficiency: The system will allow users to analyze large volumes of historical data and interpretability with the usage of data visualization tools
3. Competitive advantage: By leveraging the sales data analytics and accurate forecasting provided by the system, businesses can outperform competitors who rely on traditional prediction methods.

C. Solution Architecture

Client-Side: The web application is implemented using the Django framework.
Backend: The backend of the system is implemented using Python.
Model Integration: The stacked ensemble model for sales forecasting is integrated with the Django framework for usage through the web app as a pickle file.

V. RESULTS AND FUTURE WORKS

A. Results

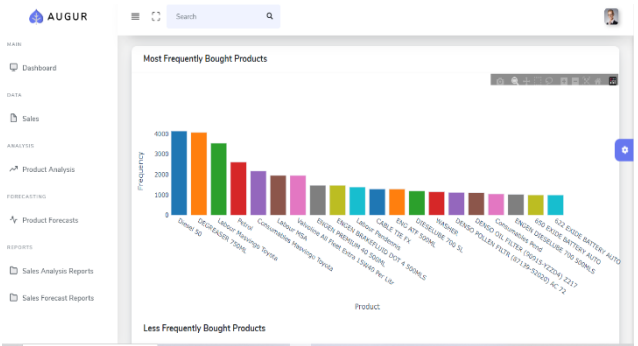

Figure 5: The interface analysis of sales data

TABLE 1: OBJECTIVES ACHIEVED

Objectives	Fully achieved	Partially achieved
To develop a web-based sales forecasting system that predicts future sales by utilizing machine learning model trained on historical data	✓	
To develop a web-based sales forecasting system that accurately forecasts sales trends over a period of time	✓	
To develop a web-based sales forecasting system that analyzes sales data	✓	

B. Future Works

Some potential areas of future work for the system could include:

1. Implementation of real-time data upload to the system to enhance forecasting accuracy and up to date data.
2. Integrating with CRM systems.
3. Usage of explainable AI to help users understand the rationale behind the sales forecasting results.

VI. CONCLUSION

In conclusion, the development of a web-based sales forecasting system using a stacked ensemble model holds immense potential for the Zimbabwean retail industry. Challenges faced by local businesses in optimizing inventory management and marketing strategies in an economically

volatile environment have highlighted the critical need for data-driven decision making. Traditional intuition-based planning methods have proved to lead to significant losses due to inefficiencies in addressing complexities of the market [12]. This system addresses the critical need for data driven insights by leveraging machine learning algorithms to generate robust and accurate sales forecasts. The system utilizes historical sales data, exchange rates and seasonality to come up with accurate sales forecasts. This integration of machine learning algorithms allows for the identification of patterns, complex relationships and seasonality in sales data which enables businesses to anticipate demand, identify growth opportunities and mitigate risks of inventory management effectively [14]. The development of a user-friendly web-application allows for the easy use of the system and the use of data visualization tools which would the easy interpretation of the results. Overally, this proposed sales forecasting system using a stacked ensemble model offers a promising solution for Zimbabwean retail businesses to thrive in an economically volatile and competitive environment.

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